

# Boosting Diary Study Outcomes with a Fine-Tuned Large Language Model

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## Abstract

This study explores fine-tuned Large Language Models (LLMs) integration into diary studies within the Human-Computer Interaction (HCI) field to enhance data collection and analysis. Leveraging a Mistral 7B model fine-tuned with a curated dataset of over 1,000 diary entries, this research addresses challenges such as participant engagement and data richness. The fine-tuned model offers personalized feedback, facilitating deeper reflection and structured recording while reducing the cognitive load on participants. The DiaryQuest educational platform, enhanced with advanced visualization tools and semantic search capabilities, enables educators to efficiently analyze diary data, extract thematic insights, and provide targeted guidance. Results from user evaluations reveal that the optimized platform improves learning outcomes, teaching efficiency, and overall user experience. By bridging traditional diary methodologies with state-of-the-art LLMs, this study advances HCI education and establishes a scalable framework for applying AI in broader educational and research contexts.

## CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Education**; • **Computing methodologies** → **Natural language processing**.

## Keywords

Diary Study, Large Language Model

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## 1 Problem and motivation

Diary study is a longitudinal research method that entails a systematic analysis of diary entries recorded over a period by participants to profoundly investigate how individuals experience and reflect on sociocultural phenomena and historical events [22]. This approach is particularly suited to exploring subtle or sporadic incidents that are difficult to capture through conventional survey methodologies [13]. In the field of HCI, the diary study method is favored for its unique perspectives and in-depth analytical capabilities and has been widely adopted to examine aspects such as how technological products impact productivity in everyday life [1, 8, 18, 24, 37], user experience in video games [15, 16, 19, 38, 39], the impact of technology on relationships [31, 44], and how children use and interact with technology [32, 36, 45]. Through meticulous analysis of diary contents, researchers can observe user behavior patterns within specific technological contexts, capture subtle behavioral details or cognitive biases that are generally unnoticed, and attempt to identify the underlying human factors for these behaviors and cognition.

Diary study, as a traditional research method, remains the challenge. Carter et al. categorized diary studies into two main types based on how participants record their experiences [3]. The first type is elicitation studies, in which participants are typically required to capture media content (such as photographs, videos, or audio) to prompt subsequent interviews. This approach has the advantage of evoking participants' memories through specific media content, thereby aiding researchers in gaining a deeper understanding of participants' experiences. However, this method may also be limited by the subjectivity of the participants in choosing what content to record, potentially leading to omissions of certain details. Li et al. designed and evaluated an automatic contextual information recording agent, DiaryHelper, at CHI 2024, which integrates contemporary LLMs and enables participants to capture rich and precise contextual information effortlessly, reducing the burden on them while enhancing the detailed recall of recorded events and offering deeper insights [30].

Inspired by DiaryHelper, our study explores the feasibility and effects of integrating LLMs with the second type of diary study, feedback diary studies. Unlike elicitation studies, feedback studies require participants to record and respond to daily life events based

on a pre-defined question structure. This method demands detailed event logging and structured presentation, providing systematic research data. However, the complexity or frequency of logging may increase cognitive load, diminish participant engagement, and consequently affect the quality and stability of research outcomes. Fan et al. proposed a framework for applying feedback diary studies in HCI education [11] and a technical support platform named DiaryQuest [46]. We realized this platform has not yet fully utilized existing LLMs to enhance participation and efficacy in diary studies. Despite the powerful natural language processing capabilities of existing LLMs, there are still certain limitations in specific scenarios [14] such as lacking depth in providing personalized feedback for complex situations [4], and there is room for improvement in guiding users to structure records with precision and engaging in human-centric interactions.

This research aims to enhance pre-trained models using fine-tuning techniques to provide personalized feedback and recommendation generation. This study intends to transform the traditional fixed question-and-answer structure into flexible, dynamic diary prompts by analyzing participants' diary entries. This customized support promotes participants' reflection and deep understanding of recorded events, enhancing their learning motivation and experience. Through developing and applying these technologies, we aim to promote the deep integration of LLMs with diary study methodologies, thereby enhancing the quality of research data and participants' learning experiences.

## 2 Background and related work

In this section, we will introduce the research background from two main aspects: first, we introduce the application of diary studies in HCI education, and second, we explain the fine-tuning techniques of large language models. The advancements in these two areas provide an essential theoretical and practical foundation for our research.

### 2.1 Diary Study In HCI Education

Previous research revealed that computing students face up to 18 learning challenges within the academic instruction of HCI [35]. These challenges include difficulties in mastering theoretical concepts and issues related to applying this knowledge to practical user experience design. Against this backdrop, Fan et al. proposed an innovative pedagogical framework (PCAR Framework) that employs diary study to enhance the effectiveness of HCI education [11]. The core of this framework is to foster student self-reflection through diary writing and to consider problems from users' perspectives, thereby improving students' profound understanding of user needs and user experience. This pedagogical framework comprises four phases: Plan, Collect, Analyze, and Reflect. This pedagogical method stimulates students to think about concerns from the user's perspective and aids them in identifying and understanding the difficulties users might encounter during actual use. Consequently, students can further contemplate improving interaction design to enhance user experience. Through this integration of reflection and practice, students can more effectively apply theoretical knowledge in practice, exploring and innovating new methods in interaction design based on personal interests and experiences.

Based on the this framework, the DiaryQuest system developed by Zhao et al. [46] represents an innovative attempt to employ diary studies in large educational settings. This system enables educators to create and manage diary assignments and track submissions of student diaries. Educators can gain deep insights into students' perspectives and discoveries by applying big data technologies to analyze diary entries. DiaryQuest offered insights into topics and sentiment trends in student diaries but faced limitations due to traditional machine learning methods like LDA [23], which produced general topics lacking nuanced context and personalization. This burdened professors and teaching assistants with providing feedback in large classrooms. Additionally, DiaryQuest struggled to maintain student enthusiasm for sustained diary writing, as long-term commitment requires significant time and effort, and maintaining content quality and depth remains a challenge. To enhance engagement and the effectiveness of diary records, this study proposes leveraging fine-tuning techniques of LLMs to provide personalized feedback and suggestions to students. This technology can offer targeted feedback, highlight strengths and areas for improvement in the diary reports, and guide students in exploring new perspectives and delving deeper into specific topics.

### 2.2 Fine-Tuning

In recent years, with the rapid development of artificial intelligence, LLMs have become a focal point of research, demonstrating significant potential and influence across various sectors, particularly in education. Fine-tuning is a training technique applied to pre-trained LLMs using the principles of transfer learning to enhance performance on specific tasks [20, 27]. The essence of this technique lies in transitioning the model from general language processing capabilities, such as understanding and generation, to handling more specialized tasks, such as sentiment analysis, legal document review, or medical report generation. According to research by Devlin et al., pre-training the BERT model captures the language's deeper structures and complex semantics. Subsequent fine-tuning allows the model to adapt quickly to new application contexts with lower computational costs without sacrificing its original language understanding and generation capabilities [9]. This not only significantly saves resources but also shortens the development cycle of models. Sun et al. indicated that fine-tuning conserves resources and generally enhances model performance, sometimes even surpassing models trained from scratch [42].

Through fine-tuning techniques, the application of LLMs in the educational sector is becoming increasingly widespread [6, 34]. These models can be utilized for automated examination scoring systems [29], providing improved explanations of programming errors [43] and assisting in developing educational curricula [26]. These applications not only optimize the allocation of educational resources but also offer new possibilities for the personalization and intelligence augmentation of education. With ongoing advancements and technological enhancements, applying LLMs in education gradually evolves from simple task automation to more profound instructional understanding and support.

### 3 Approach and uniqueness

In practice, fine-tuning typically involves several key steps: Initially, one selects a pre-trained model highly relevant to the target task as a starting point. Subsequently, a meticulously annotated dataset for the specific task is prepared, which is used to adjust the model parameters. This section will detail the specific operational procedures of this experiment, including the selection of the pre-trained model, the fine-tuning process, and the fine-tuned model outcomes.

#### 3.1 Pre-trained model Selection

To determine the most suitable LLMs as our pre-trained model, we selected six popular options: 4 open-weight models - Llama 3.1 8B, Llama 3.2 11B Vision, Gemma 2 9B, Mistral 7B, and 2 Closed-Source Reference Models - GPT-4o, GPT-4o-mini. Each model brings its unique strengths and capabilities. Llama 3.1 and Llama 3.2 are renowned for their robust performance in natural language understanding and the latter's integration of vision capabilities. Gemma 2, known for its efficiency and scalability, presents a compelling option for cost-effective deployment. Mistral 7B, while smaller, offers competitive performance, particularly in tasks that require nuanced text generation. Both GPT-4o and GPT-4o-mini, part of OpenAI's cutting-edge lineup, excel in a wide range of language tasks and offer state-of-the-art results, with the mini version providing a more accessible option for environments with limited computational resources.

To rigorously assess the performance of selected models, we used references from the paper by Chiang et al. [7] regarding the evaluation of LLMs by human preference and designed a detailed evaluation process. Initially, we manually composed 50 diary entries of varying quality, intended to encompass a diverse array of emotions and event types to ensure a comprehensive assessment. Each selected model was required to process these entries, generating corresponding summaries and feedback, with each task producing 300 results. To fairly evaluate the quality of these generated texts, we enlisted six experienced evaluators to participate in the ranking. Evaluators were instructed to independently rank each model's summaries and feedback based on the content of the original diary entries, with ranking criteria ranging from 1 to 6, where 1 represents the highest quality and 6 the lowest. To ensure the objectivity and fairness of the rankings, all evaluation materials were provided anonymously, preventing evaluators from knowing the specific source model of any summary or feedback. This method aimed to eliminate potential biases through anonymization and ensure that the evaluation results accurately reflect the performance of each model in the HCI domain.

To synthesize these individual rankings into a consolidated outcome, the Borda count method [28] was utilized. This method assigns points to rankings in a manner that inversely correlates with their numerical value—higher rankings receive more points. These points were then accumulated for each LLM across all entries and raters to derive the total Borda scores. Analysis of these scores revealed that GPT-4o outperformed the others, demonstrating the highest overall efficacy. It was closely followed by Mistral 7B and GPT-4o-mini, which exhibited similarly robust performances (See Table 1). Further, in examining the consistency of ratings among

**Table 1: Total Borda Scores for Each LLM**

| Model                | Total Borda Score |
|----------------------|-------------------|
| GPT-4o               | 1308.0            |
| Mistral 7B           | 1127.0            |
| GPT-4o-mini          | 1101.0            |
| Llama 3.1 8B         | 1010.0            |
| Llama 3.2 11B Vision | 896.0             |
| Gemma 2 9B           | 840.0             |

**Table 2: Kendall's Tau Coefficients for Evaluator Agreement**

| Evaluator   | Kendall's Tau | p-value |
|-------------|---------------|---------|
| Evaluator 1 | 0.333         | 0.469   |
| Evaluator 2 | 0.733         | 0.056   |
| Evaluator 3 | 0.276         | 0.444   |
| Evaluator 4 | 0.828         | 0.022   |
| Evaluator 5 | 0.600         | 0.136   |
| Evaluator 6 | 0.867         | 0.017   |

different evaluators, we utilized Kendall's Tau coefficients to quantitatively measure the concordance between individual evaluator scores and the aggregated ranking from all evaluators (See Table 2). Kendall's Tau, a non-parametric statistic, is particularly effective in assessing ordinal relationships and offers a reliable measure of correlation without assuming a normal distribution of the data [40]. This approach is essential to ascertain the degree to which individual assessments conform to the collective judgment, thereby validating the outcomes of the rankings in our study. This methodological choice enhances the robustness of our findings, ensuring that the rankings reflect a true consensus among the raters rather than anomalies in individual assessments.

#### 3.2 Fine-tuning Process

After a detailed evaluation of summarization and feedback generation on six LLMs, we concluded that the open-weight Mistral 7B performed second best after our close-sourced reference model GPT-4o was an ideal choice for our work. Given Mistral 7B's strong performance metrics, resource consumption, and applicability of various tasks, we selected it as our base model. This ensures the highest efficiency and optimal results in subsequent development and deployment phases.

Over the past two years, Virginia Tech has employed the diary study pedagogical approach in its undergraduate Introduction to HCI course. During this period, students conducted comprehensive studies on 19 different mobile and web platform applications spanning multiple domains. Throughout this process, students amassed a total of 1,372 diary entries. Following preliminary organization and screening of the diary entries, we excluded those with evident errors and flaws, ultimately retaining 1046 entries. Each entry contains over 500 words of detailed description, encompassing aspects such as user experience, user emotions, and changes in user behavior. These diary entries will serve as the data foundation for this

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diary Entry:</b><br>After spending some time with this app, I've discovered that adding images to certain affirmations is pretty helpful. I was not aware of this feature since I thought this app was only used for people to write down their affirmations. Images can be helpful when it comes to putting a visualization of goals I set.                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Base Model (Mistral 7B) Feedback:</b><br>Great observation! Utilizing images to accompany affirmations is a powerful tool for visualization and reinforcement of your goals. This feature can significantly enhance the effectiveness of your affirmations by providing a tangible representation of what you're striving for. Keep exploring and experimenting with different images to see which ones resonate most with your affirmations. Additionally, consider reflecting on how often you're using this feature and if there are any ways you can incorporate it more consistently into your routine. | <b>Fine-tuned Mistral 7B Feedback:</b><br>This diary entry effectively conveys your discovery of a useful feature in the app, highlighting the benefit of adding images to affirmations for enhancing visualization. This adds a practical dimension to your reflection, showcasing your engagement with the app and openness to its capabilities. To improve the entry, consider expanding on how the images specifically aid in visualizing your goals. Sharing any particular images you find effective or any specific goals they help visualize could provide deeper insight into your |

(a) Fine-tuned Mistral 7B &amp; Base Model Feedback Comparison

**Strengths:**

- The entry includes specific details like the environment and weather conditions, which add context to your user experience.
- You noted the speed of the rewards, indicating an awareness of system performance and user satisfaction.

**Areas for Improvement:**

- Expand on your personal feelings or thoughts about the interaction, such as your surprise or confusion regarding the speed of rewards. This can provide deeper insight into your user experience.
- Consider analyzing the user interface or design aspects that might have contributed to the experience, like the effectiveness of the scanning animation. This could offer a more comprehensive analysis.
- Elaborate on the role of other players' comments in your experience and how they influenced your perception of the app.

(b) Personalized Feedback

**Figure 1: Fine-tuned Mistral 7B Model Outputs**

study. It should be noted that previous academic studies have repeatedly demonstrated the effectiveness of supervised fine-tuning on a small number of high-quality samples [5, 47]. This suggests that even datasets of limited size, provided they are of sufficient quality, can yield effective learning outcomes through precise training methodologies. Consequently, this study opts to conduct an in-depth analysis and learning based on these 1046 selected diary studies, aiming to promote their broader application in practical contexts.

To ensure data privacy and security, we leveraged SpaCy's EntityRecognizer to remove sensitive information such as names, contact details, and personal identity information during preprocessing [41], effectively reducing the risk of potential data leakage. Additionally, to improve the consistency and accuracy of data processing, we formatted the data entries to ensure standardization between different data sources and types. Furthermore, we eliminated all redundant data unrelated to diary study content analysis, such as formatting tags and non-textual content within the documents. This not only cleansed the dataset but also optimized the subsequent data handling and analysis processes. Through these comprehensive preprocessing measures, we ensured the data security and the analysis's effectiveness, providing a high-quality data foundation for the research.

The fine-tuning process involved creating a triad of dataset structure. This structured triad was crucial for instructional tuning of the Mistral 7B model for tasks in the diary study. The methodology followed a knowledge distillation approach [25], using the responses of a larger LLM with over 70B parameters as high-quality labels (teacher model), which were meticulously reviewed and edited by human evaluators to ensure precision and relevance before supervising the Mistral 7B (student model) training. As a result, Mistral 7B maintained robust response capabilities in summarization and feedback generation tasks while using fewer parameters than its teacher model, ensuring both efficiency and high performance.

We integrated Low-Rank Adaptation (LoRA) layers into Mistral 7B to preserve most of the pre-trained parameters while introducing small, low-rank updates ( $BA$ ). Specifically, LoRA only modified the attention weight matrices ( $W_q$  and  $W_o$ ) that are important for capturing context during the self-attention process [21].

- **Parameter-Freezing:** LoRA reduces the computational load by freezing most of the weights of the Mistral 7B model during fine-tuning.
- **Low-Rank Updates:** Task-specific adaptation is parameterized by low-rank updates  $\Delta W = BA$ . Here,  $B \in \mathbb{R}^{d \times r}$  and  $A \in \mathbb{R}^{r \times k}$ , where  $r \ll \min(d, k)$ . This approach significantly reduces the number of learnable parameters while maintaining model performance.
- **Integration with Transformer Layers:** LoRA applies the self-attention mechanism of the Transformer architecture to the key weight matrices ( $W_q$  and  $W_o$ ). These layers have the greatest impact on adapting to task-specific contexts.
- **Reparametrization:** During the forward passes, the model combines the pre-trained weights  $W_0$  and the LoRA updates  $\Delta W : h = W_0x + \Delta Wx = W_0x + BAx$ . [21]

The training details for the fine-tuning process of the Mistral 7B model include the use of 13,222,350 training tokens across 10 epochs. The learning rate was set at 0.0001, which facilitated the effective adaptation of the model to the specific requirements of diary entry summarization and feedback generation. By fine-tuning cutting-edge technology and tailoring it to the specific goals of the platform, the Mistral 7B model has been optimized to enable efficient and effective summarization and feedback generation for diary entries (See Figure 1).

### 3.3 DiaryQuest Data Analysis and Visualization Features

We introduce data analytics and visualization features based on fine-tuned LLM into the DiaryQuest platform to support diary studies in educational settings effectively (See Figure 2). By converting student-submitted diary texts into vector representations and indexing and clustering them within a FAISS vector database [10], the system can rapidly perform similarity searches, topic extraction, and visualization for large volumes of text data. In particular, for topic extraction, we combine the K-Means clustering algorithm [17] with the fine-tuned Mistral 7B model to group diary entries without human intervention and distill core themes. This approach enables educators to quickly gain insights into different student cohorts' learning experiences and potential issues.

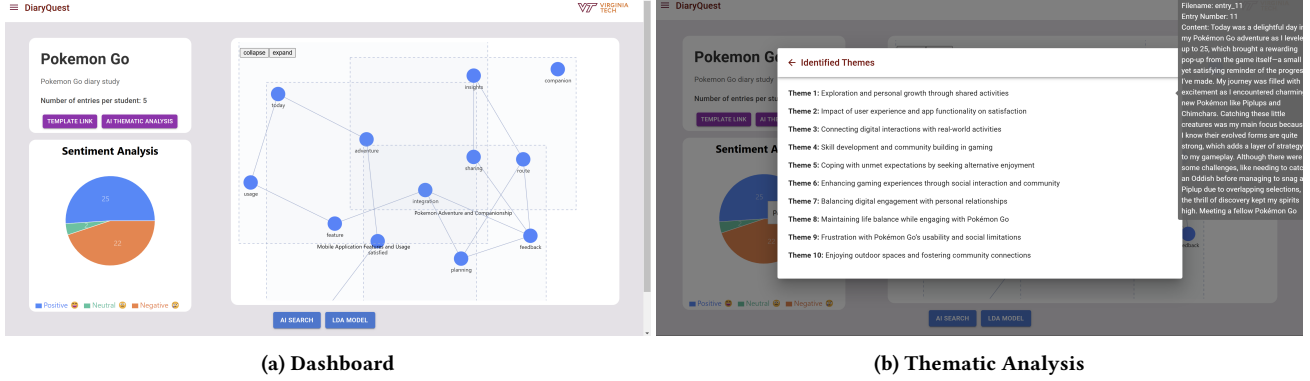


Figure 2: Data Analysis and Visualization Features

We offer visualizations of clustered themes and incorporate semantic search and semantic graph modules to further enhance data exploitability. Semantic search allows users to query the FAISS index with natural language or keywords, followed by automatic summarization and citation labeling generated by Mistral 7B. This ensures users can accurately locate relevant content in a high-dimensional semantic space. Meanwhile, semantic graph visualization, leveraging LDA topic modeling and Doc2Vec similarity analysis [2], maps thematically similar diary entries as connected nodes, which affords a comprehensive view of frequently occurring themes and potential relationships and difference. DiaryQuest also integrates sentiment analysis modules to enhance the interactive experience. The sentiment analysis module employs lightweight models such as DistilBERT [33] to automatically label diary entries as positive, negative, or neutral and visually represent the overall sentiment distribution. Users can click on each sentiment category through an intuitive interface to view the associated diary entries, thereby gaining deeper insights into students’ unique thought processes.

The system offers instructors a global view of diary analyses and visualizations, facilitating instructional reflection and decision-making from overall trends or class-level perspectives. For students, the platform displays their diaries and corresponding AI-generated feedback and provides various data visualization tools that enable them to interpret analysis results. These tools deepen students’ understanding of their learning habits and performance and support personalized learning path planning. Students can identify and strengthen their areas of expertise and interest in learning by selecting appropriate data perspectives and analysis methods based on their interests and learning needs. Moreover, DiaryQuest encourages diverse forms of interaction among students. Students can connect with peers with similar interests or learning objectives by sharing their diaries and feedback. Such interaction goes beyond mere exchanges about diary content, encompassing discussions of data interpretation and collaborative exploration of questions and solutions.

## 4 Results and contribution

In order to conduct an in-depth evaluation of our fine-tuned Mistral 7B model, we designed a survey focusing on the assessment of the

quality and effectiveness of the feedback generated by the fine-tuned model and user satisfaction following the introduction of the fine-tuned model features on the diary study platform DiaryQuest. A total of 12 participants (2 instructors & 10 students) were involved in this questionnaire survey.

| Quality                  | Avg  | Std Dev | Min | Max |
|--------------------------|------|---------|-----|-----|
| Detailed (500+ Words)    | 3.92 | 0.90    | 2.0 | 5.0 |
| Ordinary (200-300 Words) | 4.33 | 0.78    | 3.0 | 5.0 |
| Brief—into (100 Words)   | 4.00 | 0.74    | 3.0 | 5.0 |

Table 3: Participants Rating on Diary Feedback from Fine-tuned Model

Participants rated their overall satisfaction from several key dimensions: ease of use, feature completeness, and user interface friendliness. The results indicated that users highly rated the optimized DiaryQuest platform, with an average score of 4.58 out of 5. Participants particularly praised how these new features helped them more efficiently record and analyze daily data and present this information through intuitive charts and reports, allowing for deeper analysis and understanding of their activities and habits.

We input diary entries of three different quality levels—detailed, ordinary, and brief—into the fine-tuned Mistral 7B model. Participants employed a 5-point rating system (1 being the lowest and 5 the highest) to evaluate the feedback generated by the model. Our fine-tuned Mistral 7B model received relatively high ratings (See Table 3). The fine-tuned model demonstrates commendable adaptability and targeted effectiveness for both brief and ordinary diary entries. Through an integrated analysis of text content, emotional orientation, and writing style, the model can provide constructive criticism and suggestions for improvement, thereby assisting users in enhancing their writing skills. Furthermore, the model can offer personalized feedback based on key information within the diary entries, aligning each writing experience with the user’s personal growth needs. This intelligent feedback mechanism increases the enjoyment of writing and significantly promotes the rapid development of users’ writing capabilities. However, for information-rich detailed diary entries, the feedback generated by the model may not offer much additional value. In processing such entries, the

model primarily focuses on organizing and analyzing existing content to confirm consistency and logic but struggles to enhance its richness or depth. Future research should explore optimal times and methods for AI interventions in education to maximize their educational impact [12].

In this study, we integrated the fine-tuned Mistral 7B LLM into a diary study platform, DairyQuest, achieving automated analysis and personalized feedback on students' diary entries. This innovative approach enhanced the platform's accuracy and depth in text comprehension and information mining and altered the traditional interaction mode between teachers and students within the diary study model. Specifically, upon submitting their diaries, students could instantly receive intelligent feedback based on their learning status and reflective focus, thereby stimulating more profound thought and discussion. Furthermore, using LLM-driven thematic analysis and innovative search functions significantly reduced the workload of teachers and teaching assistants in diary grading and evaluation. This allowed them to devote more energy to providing personalized guidance and improving course quality, which enhanced teaching efficiency and improved the quality of education.

From a broader perspective, this study offers meaningful insights and tools for the HCI field. By integrating semantic graph technology and advanced search mechanisms, the research establishes a novel paradigm for text analysis. This enables researchers and developers to extract themes, associative relations, and latent patterns from large-scale unstructured texts more efficiently. This integrated approach applies not only to extensive diary study data analysis but can also be extended to other educational settings and a broader range of text analysis domains, thereby laying a solid foundation for the deepened application of HCI technologies in fields such as education and social science research.

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